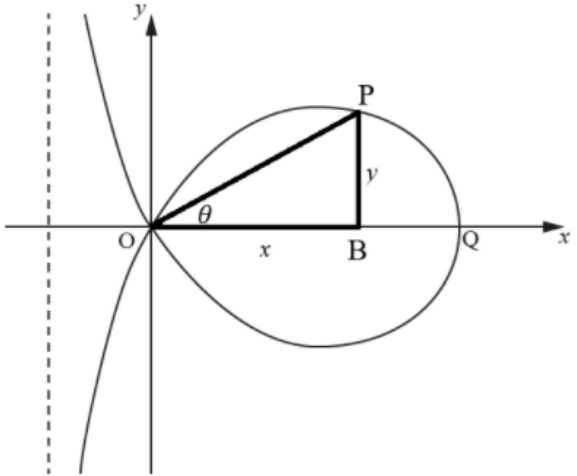
Question			Answer	Marks	AO	Guidance
14	(a)			B1 [1]	3.1a	• Triangle POB drawn, where B is the point on the x- axis such that PB is perpendicular to the x-axis. • P may be anywhere on curve in 1st quadrant. • The points P and B need not be labelled • Vertical and horizontal sides should be labelled y and x OR x and y clearly indicated on the axes OR Point on curve labelled (x, y) • Allow angle θ not labelled • Condone good freehand – mark intent
14	(b)		$(3a - x =)\ 3a - \dfrac{a(3-t^2)}{1+t^2}$ $= \dfrac{3a + 3at^2 - 3a + at^2}{1+t^2}$ $= \dfrac{4at^2}{1+t^2}$	B1 [1]	2.1	AG • Correct completion to given result. • Must see at least one step of working and no errors or omissions
14	(c)		$(a + x =)\ \dfrac{4a}{1+t^2}$	B1 [1]	3.1a	• Correct expression in terms of a and t • Must be simplified

Question			Answer	Marks	AO	Guidance
14	(d)		$\frac{3a-x}{a+x} = \frac{\frac{4at^2}{1+t^2}}{\frac{4a}{1+t^2}}$	M1	3.1a	• Dividing the correct expressions for $3a-x$ and $a+x$
			$= t^2$ $\Rightarrow \frac{3a-x}{a+x} = \frac{y^2}{x^2}$ $\Rightarrow \frac{x^2(3a-x)}{a+x} = y^2$	A1	2.1	AG • Simplifying to t^2 • Using $t = \frac{y}{x}$ to give a correct completion to given result • Must have at least one line of working and no incorrect work seen
			Alternative method 1 $\frac{x^2(3a-x)}{a+x} = \frac{x^2 \times \frac{4at^2}{1+t^2}}{\frac{4a}{1+t^2}}$	M1		• Substituting correct expressions for $3a-x$ and $a+x$ into $y^2 = \frac{x^2(3a-x)}{a+x}$
			$= \frac{x^2 \times t^2}{1}$ $= y^2$ (since $t = \frac{y}{x}$)	A1		AG • Simplifying to x^2t^2 • Using $t = \frac{y}{x}$ to give a correct completion to given result • Must have at least one line of working and no incorrect work seen

			Alternative method 2 $\frac{x^2(3a-x)}{a+x} = \left(\frac{a(3-t^2)}{1+t^2}\right)^2 \times \frac{1}{a+\frac{a(3-t^2)}{1+t^2}} \times \frac{4at^2}{1+t^2}$	**M1**		• Substituting $x = \frac{a(3-t^2)}{1+t^2}$ and $3a - x = \frac{4at^2}{1+t^2}$ into $y^2 = \frac{x^2(3a-x)}{a+x}$
			$= \frac{a^2(3-t^2)^2}{(1+t^2)^2} \times \frac{4at^2}{a(1+t^2)+a(3-t^2)}$ $= \frac{a^2(3-t^2)^2}{(1+t^2)^2} \times \frac{4t^2}{1+t^2+3-t^2}$ $= \frac{a^2(3-t^2)^2}{(1+t^2)^2} \times \frac{4t^2}{4} = \frac{a^2(3-t^2)^2t^2}{(1+t^2)^2}$ $= \left(\frac{a(3-t^2)t}{(1+t^2)}\right)^2 = y^2$	**A1**		AG • Simplifying and showing $y^2 = (xt)^2$ to give a correct completion to given result • Must have at least one line of working and no incorrect work seen
			Alternative method 3 $x = \frac{a(3-t^2)}{1+t^2}$ $\Rightarrow x + xt^2 = 3a - at^2$ $\Rightarrow t^2(x+a) = 3a - x$ $\Rightarrow t^2 = \frac{3a-x}{a+x}$	**M1**		• Making t^2 the subject of $x = \frac{a(3-t^2)}{1+t^2}$ as far as $t^2 = \ldots$ • Condone algebraic or sign slips
			$y = xt \Rightarrow y^2 = x^2t^2 = \frac{x^2(3a-x)}{a+x}$	**A1**		AG • Using $t = \frac{y}{x}$ to give a correct completion to given result • No incorrect work seen
				[2]		